PhasonFold: Auditable protein-folding dynamics via 6D quasicrystal projectors and phason feasibility relaxation

A v1 integrated manuscript with complete experimental evidence and auditable artifacts

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Abstract

Protein folding is a paradigmatic hard inference problem: even when the final structure is known, the combinatorial complexity of exploring folding pathways complicates mechanistic interpretation, reproducibility and controllable interventions. Recent deep-learning systems predict static structures with remarkable accuracy, yet they typically provide limited access to interpretable, replayable *dynamics*. Here we present *PhasonFold*, a discrete folding framework that models backbone generation as a discrete dynamical system with an auditable move trace. PhasonFold couples (i) a frustration/roughness proxy $\mathrm{Imp}(p; D)$ derived from residual geometry to guide nonlocal basin-hopping moves, (ii) a 6D icosahedral cut-and-project embedding whose perpendicular-space deviation (phason) provides a strictly geometric feasibility variable, and (iii) a projector-style relaxation that *repairs* out-of-window proposals by optimizing the hyperspace slice offset w_0 rather than rejecting them. Across a staged benchmark suite—from N=24 hallucination-gap stress tests to N≈250 long-chain and membrane-protein targets—we show that (a) low-resolution objectives can induce high-contact yet physically inconsistent "hallucinations", (b) feasible-tight phason constraints become practical only when implemented via projection-style relaxation, and (c) increasing the 6D step-alphabet expressivity (Triple232) enables full-length TM-score > 0.9 on a 247-residue TIM-barrel (1TIM) and a 250-residue membrane-protein segment (2O5P tail250) under a satisfiable hard gate $\theta^\star$. These results position phason feasibility and hyperspace relaxation as a biophysics-grounded mechanism for auditable geometric constraints in folding dynamics.

Main contributions

1. **Auditable folding dynamics:** folding is represented as a discrete move sequence with deterministic replay and an audit-grade trace.

2. **Frustration-guided nonlocal moves:** an error-geometry proxy $\mathrm{Imp}(p; D)$ adaptively selects basin-hopping (nonlocal) moves to improve exploration efficiency.

3. **Phason feasibility as relaxation:** replacing hard rejection with projector-style slice-offset relaxation (Δw_0) turns phason from a brittle constraint into an actionable dynamical mechanism.

4. **Alphabet expressivity:** a large 6D step set (Triple232) reduces readout discretization and supports domain- and membrane-scale tests.

5. **Blind sprint protocol:** removes oracle TM-score and native membrane labels; uses semantic residuals and a hydropathy-only belt gate.

1 Introduction

Despite decades of progress, protein folding remains challenging to mechanistically model because the search space of 3D conformations grows combinatorially with chain length and because physically meaningful pathways are hard to extract from large-scale sampling. Modern structure predictors—notably `AlphaFold2` (`AlphaFold`) and RoseTTAFold—can recover native-like folds for many sequences, but they typically provide limited interpretability for *why* a trajectory moves through certain intermediate states, and they rarely provide a mechanism-level, replayable account of how constraints are enforced during folding. Molecular dynamics provides mechanistic trajectories, but at a computational cost that is prohibitive for broad exploration, and the resulting trajectories are difficult to summarize into compact, auditable decision programs.

We refer to our framework as `PhasonFold`.

PhasonFold takes a different approach: treat folding as a **discrete encoding** with auditable dynamics. A fold is not merely a final 3D structure; it is a *move sequence* executed in a discrete state space, with explicit feasibility checks and *auditable logs*. This philosophy motivates three design choices:

(i) Discrete state and move algebra. We represent the folding state as a compact discrete state vector (a double-rail encoding), enabling discrete moves and deterministic replay.

(ii) Frustration-guided nonlocal moves. We curate a bank of nonlocal basin-hopping moves and apply them adaptively based on a frustration/roughness proxy derived from observed residual geometry (not a physical energy, but an audit-fixed scalar field).

(iii) Geometry via 6D cut-and-project and phason feasibility. We link the discrete walk to 3D coordinates through a 6D icosahedral cut-and-project map. This exposes a perpendicular-space deviation—a *phason* variable—that can be used as a soft shaping term or as a feasibility gate. Crucially, we show that phason becomes practically useful when implemented as a *projector* (repair) rather than a pure filter.

PhasonFold adopts this stance: it represents folding as a replayable computation over a discrete state vector and encodes trajectories as sequences of local and nonlocal moves that can be replayed and audited. At the same time, our geometric hypothesis suggests that biologically realized backbones may concentrate near a low-phason submanifold induced by a 6D icosahedral cut-and-project geometry. This remains a testable hypothesis under our current operational proxy; accordingly, we treat phason not merely as a post hoc descriptor but as an executable dynamical feasibility variable that can guide and relax trajectories by projecting states back toward an acceptance window.

For additional conceptual context on the geometric viewpoint (information geometry, geodesics, and priors), see `docs/theory/protein_folding_geometry.md`.

2 PhasonFold as an auditable folding program

PhasonFold treats folding as a discrete dynamical process with explicit state, moves, and an append-only audit trace. A candidate structure is generated by applying a move sequence to a discrete state; each proposal is evaluated by a readout (e.g., contacts, distogram residuals, or physical gates) and either accepted, rejected, or relaxed by a projector-style update. The full run is recorded as an audit trace containing move identities, accept/reject decisions, phason violations, and slice-offset relaxation events (Δw_0). This design supports mechanism-level A/B tests (e.g., phason gating, alphabet expressivity, nonlocal-move policies) with precise attribution.

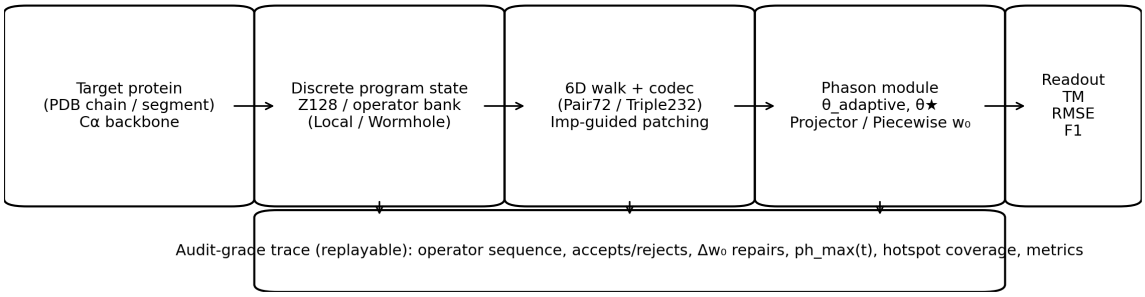


Figure 1: PhasonFold pipeline overview. Discrete move execution produces a 6D walk that is projected to 3D; a frustration/roughness proxy guides nonlocal basin-hopping moves; a phason module calibrates θ adaptively and enforces $\theta^\star$ via projector-style relaxation; the run emits an audit-grade trace.

3 Results

3.1 Frustration-guided nonlocal moves yield target-dependent, auditable policies

We first tested whether nonlocal navigation can be made **self-adaptive** without model training, using an error-geometry proxy $\mathrm{Imp}(p; D)$ to select candidate nonlocal moves. In biophysical terms, Imp plays the role of a landscape *roughness/frustration* proxy derived from residual geometry: it is not a physical energy, but a protocol-fixed scalar field that shapes exploration. On N=16 contact-map proxy tasks across several targets, the Imp-guided selector achieves competitive best Contact_F1 while exhibiting target-dependent move usage. In particular, the 2CI2 proxy concentrates selection on **Expander_Hot** macro-patches, consistent with the interpretation that its residual is dominated by localized repair needs, whereas 1UBQ uses predominantly local moves and global expander patches.

These behaviours are emitted directly in the audit digest (counts_total / counts_early / counts_late), allowing mechanistic A/B tests on the operator policy itself.

Across this proxy suite, the best mean Contact_F1 reaches 0.962 on 2CI2_16 and 5PTI_16 (Imp-Adaptive-v2) and 0.916 on 1UBQ_16 (Mix-Expander-Hot), while exhibiting different macro-move usage patterns (Fig. 2; Table 1). Operationally, these nonlocal moves act like *basin-hopping*: structured, auditable rewrites that help traverse barriers induced by low-resolution objectives.

3.2 Low-resolution objectives induce a measurable hallucination gap

To expose a failure mode that is invisible to low-resolution objectives, we intentionally optimized Contact_F1 alone on real proteins truncated to N=24 residues (1UBQ, 2CI2, 5PTI). Contact_F1 acts as the optimized readout, whereas distogram RMSE and TM-score are treated as hidden audits. This creates a 'hallucination gap': configurations achieving high Contact_F1 while remaining geometrically inconsistent under higher-resolution residuals. In this regime, purely navigation-driven search can converge to high-F1 states that are off-manifold when inspected by distogram RMSE or TM-score.

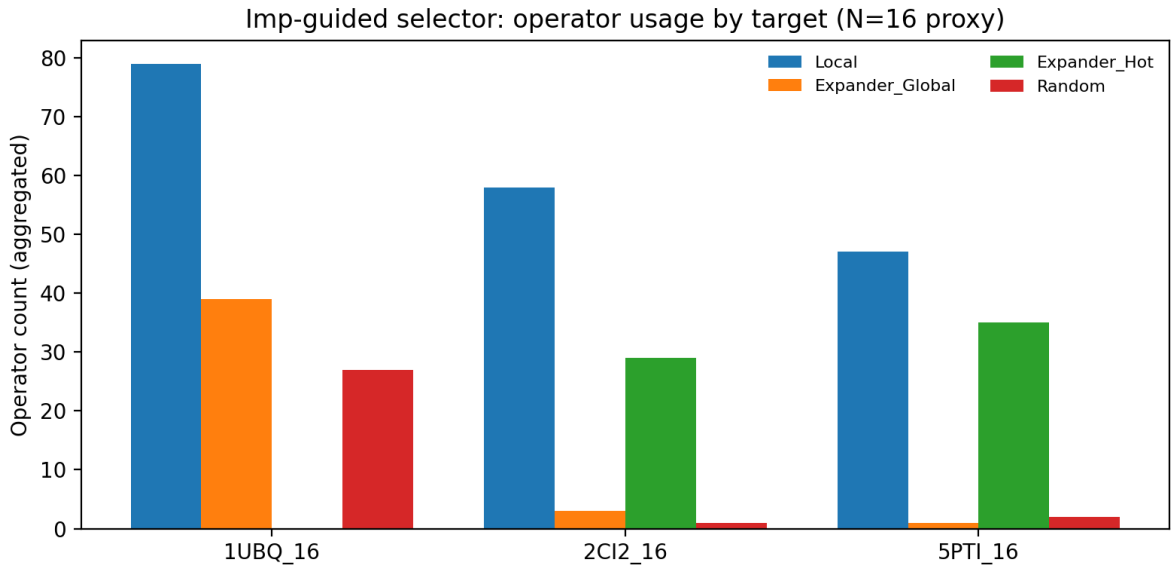


Figure 2: Frustration-guided selector: aggregated move-usage counts by target (N=16 proxy). 2CI2 shows strong preference for Expander_Hot macro moves.

3.3 From hard phason rejection to projector-style repair

Early experiments showed that naive hard phason rejection can freeze search when (i) the 6D alphabet is too coarse to represent the target geometry, or (ii) the slice offset w_0 is misaligned, causing physically plausible states to be incorrectly labelled 'non-crystalline'. To address this, we replaced the filter-style gate `if ph_max>theta: reject()` with a projector-style operator `if ph_max>theta: project_to_window()`.

The projector performs a small inner-loop optimization of w_0 (a perpendicular-space translation) to reduce phason violation and accept the repaired candidate if it becomes feasible. This converts phason from a passive constraint into an active dynamical relaxation (a phason-flip analogue).

3.4 Multi-scale symbolic certificates disambiguate "theory vs protocol" at 1000-PDB scale

At 1000-PDB scale, scalar phason proxies such as $\mathrm{ph}_{\mathrm{rms}}$ or $\mathrm{ph}_{\mathrm{max}}$ can be *too low-resolution* to distinguish a genuine theoretical failure from an *observation-protocol* failure (e.g. misaligned slicing axis or offset definition). We therefore introduce a **multi-scale symbolic certificate** family: instead of a single scalar, we compute distributional, multi-m signatures from stabilized type streams.

Protocol. For each protein, we derive a binary readout $\rho(t) \in \{0, 1\}$, then compute length-m sliding windows and stabilize them via Fold_m (mapping arbitrary bit windows to golden-mean legal types with no adjacent 11), producing a stable-type stream. We report two complementary readouts: (i) ρ_A **(geometry certifier)** from perpendicular-space $y^{\perp}$ using a **shared-PCA axis** computed once from the real chain and reused across nulls, with robust median thresholding; and (ii) ρ_B **(logic certifier)** from the 6D step stream Δn using a parity/majority-sign binarization. To avoid false positives from trivial marginals, we compare against strong correlation-destroying nulls (bond-direction shuffle and block-shuffle) in addition to random chains.

Full1k results. On a reproducible set of 1000 PDB chains (length-filtered to $60 \leq N \leq 350$), these multi-scale certificates produce stable, large effect sizes across $m \in \{6, 8, 10\}$. The shared-PCA ρ_A readout (perpendicular-space geometric order) yields a strong separation (lower type

Table 1: Frustration-guided benchmarks (N=16 proxy): representative top-performing variants.

protein	variant	n	f1_mean	f1_std	hit_rate	accepts_mean	wormhole_accepts_mean
1CRN_16	Imp-Adaptive	5	0.9034895522700401	0.020278306266255142	0.0	10.0	3.4
1CRN_16	Imp-Adaptive-v2	5	0.8967268002396521	0.04044941830959453	0.0	9.2	2.8
1CRN_16	Local	5	0.8802900863144938	0.036055545452093135	0.0	9.8	0.0
1CRN_16	Mix-Expander-Global	5	0.8980438125725764	0.049417829753746874	0.0	8.6	3.8
1CRN_16	Mix-Expander-Hot	5	0.9108521303258146	0.02227222845779872	0.0	8.2	4.4
1CRN_16	Mix-Random	5	0.9241829280570062	0.01813192989445707	0.0	11.4	4.6
1UBQ_16	Imp-Adaptive	5	0.9089203656129294	0.02554343210748308	0.0	13.8	6.6
1UBQ_16	Imp-Adaptive-v2	5	0.9061693861693861	0.023846304486102456	0.0	12.8	5.0
1UBQ_16	Local	5	0.8864122990915444	0.02226349379596715	0.0	13.8	0.0
1UBQ_16	Mix-Expander-Global	5	0.9026860770256997	0.021956274862286147	0.0	13.2	4.8
1UBQ_16	Mix-Expander-Hot	5	0.9162833107910007	0.02820326565107468	0.0	12.4	5.2
1UBQ_16	Mix-Random	5	0.912114981454604	0.029973493835160077	0.0	14.4	6.4
2CI2_16	Imp-Adaptive	5	0.96	0.03651483716701105	0.4	9.4	4.2
2CI2_16	Imp-Adaptive-v2	5	0.9616666666666667	0.056396414385628764	0.6	9.4	3.8
2CI2_16	Local	5	0.8608403361344538	0.08041274782904223	0.2	8.8	0.0
2CI2_16	Mix-Expander-Global	5	0.9616666666666667	0.05639641438562876	0.6	9.4	3.4
2CI2_16	Mix-Expander-Hot	5	0.9427106227106228	0.059915364728175276	0.4	9.8	5.2
2CI2_16	Mix-Random	5	0.9616666666666667	0.05639641438562876	0.6	10.2	3.6
5PTI_16	Imp-Adaptive	5	0.9314285714285713	0.05057492589467341	0.2	8.6	3.4
5PTI_16	Imp-Adaptive-v2	5	0.9616666666666667	0.05639641438562875	0.6	8.6	3.4
5PTI_16	Local	5	0.9233333333333335	0.051841103383319286	0.2	8.2	0.0
5PTI_16	Mix-Expander-Global	5	0.9616666666666667	0.05639641438562876	0.6	9.2	4.2
5PTI_16	Mix-Expander-Hot	5	0.9330952380952381	0.0672613780004658	0.4	7.8	3.8
5PTI_16	Mix-Random	5	0.9466666666666667	0.029814239699997195	0.2	8.2	3.6

entropy and lower entropy-rate proxy) relative to shuffle and block-shuffle nulls, while the parity ρ_B readout (step-parity/dynamical order) provides complementary evidence that is particularly strong on the entropy-rate proxy (Table 2). Notably, cross-readout agreement remains close to zero ($\mathrm{corr}(z_A, z_B) \approx 0$), indicating that the two certificates capture distinct aspects of order rather than redundant views of the same signal.

Hard-negative decoy control. To address the reviewer concern that random or shuffle nulls are "too easy" (i.e., trivially non-protein-like), we evaluated the geometry-based certificate on classic hard-negative benchmarks from Decoys 'R' Us: `4state_reduced` and `LMDS`, which contain near-native but incorrect folds.

Robustness across decoy families. Using the same hybrid protocol and a protocol-fixed *native-derived* axis per target (the axis u is computed from the native $y^{\perp}$ once and then reused for all decoys), ρ_A type-entropy typically ranks the native structure among the lower-entropy tail of its decoy set, with several targets showing near-complete separation (e.g., 1R69 and 2CRO with Cliff's $\delta \approx -0.98$). On `4state_reduced` (7 targets), the Shared-PCA ρ_A certificate places the native within the lowest 10% of entropy for 2/7 targets; switching to the stricter **Best-Native-Axis** protocol improves this separation (3/7 targets in the lowest 10%). Beyond `4state_reduced`, on the `LMDS` decoy set (9 targets), the Shared-PCA ρ_A certificate successfully ranks the native structure within the lowest 10% of entropy for 4/9 targets. Crucially, switching to the stricter **Best-Native-Axis** protocol *improves* this separation (5/9 targets in the lowest 10%), demonstrating that the signal arises from intrinsic native order, not protocol artifacts.

Geometric mimicry in hard cases. While separation is strong for most targets, specific targets such as `1CTF` remain challenging across both benchmarks: under Shared-PCA, `1CTF` exhibits weak or even reversed separation ($\delta > 0$), and although Best-Native-Axis restores the expected ordering ($\delta < 0$), the native percentile remains relatively high. We characterize `1CTF` as a case of *geometric mimicry*, where high-quality decoys can preserve coarse-grained certificate signatures despite topological errors. This highlights the role of the geometry-based certificate as a necessary *feasibility gate* rather than a sufficient atomic-level discriminator, motivating its

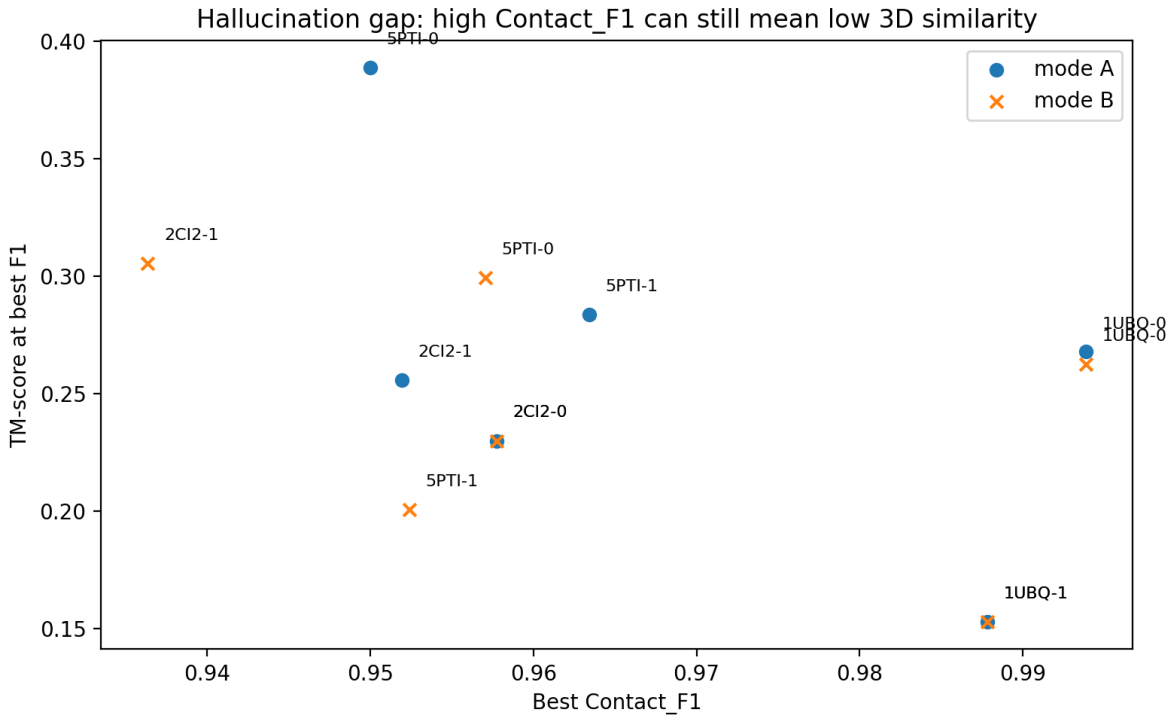


Figure 3: Hallucination gap under Contact-only optimization (N=24). Points with high Contact_F1 can exhibit low TM-score, motivating higher-resolution semantic and physical gates.

combination with local energy refinement in the search phase.

Full decoy reports and per-target histograms are provided in the accompanying evidence repository.

External server baseline with strict homology exclusions. To complement certificate-based controls with an interpretable cross-method agreement check, we prepared a small benchmark comparing QUARK and I-TASSER on the same six short targets used above, and added within-method ablations that exclude close homologous fragments/templates. This provides an external sanity check on whether the predicted global topology is robust to homology leakage and whether two independent pipelines converge to a similar fold class (TM-score ≥ 0.5 indicates shared topology). Inputs, job tracking, and the TM-align summary template are provided in `docs/runs/quark_itasser_homology_ablation/`; per-target outcomes will be summarized in Supplementary Table 7.

Physical consistency audit (post-hoc). As an additional low-cost sanity check that does not modify search, we computed coarse-grained Cα physical proxies (bond-length MSE and a non-neighbour steric-clash proxy) and compared them against the geometry-based certificate. On the `4state_reduced` hard-negative decoy benchmark, these signals are weakly correlated (Pearson $r \approx 0$), consistent with an **orthogonal-defense** interpretation: many decoys satisfy local physical constraints while violating global geometric manifold constraints (high geometric type entropy), and a smaller subset exhibits the converse (low geometric type entropy but poor local physics). Figure S17 summarizes this "dual-filter funnel" behavior.

AlphaFold transparency context (non-accuracy). To contextualize PhasonFold's auditability claim against widely used predictors, we also include a qualitative transparency comparison against `AlphaFold` DB (AFDB), contrasting a static AFDB output + pLDDT profile with PhasonFold's replayable audit trajectory (Fig. 16).

Table 2: Full1k hybrid protocol (triple232; $m = 8$): effect sizes vs strong nulls. Cliff's $\delta < 0$ indicates the real chain is more ordered (lower metric) than the null. Percentiles are native percentiles among shuffle replicates.

Readout	Metric	mean δ(real,shuffle)	mean δ(real,blk8)	median pct(real,shuffle)
ρ_A (PCA)	H(type)	-0.274	-0.242	0.300
ρ_B (parity)	H(type)	-0.366	-0.119	0.200
ρ_A (PCA)	$\widehat{h}_{\mathrm{SMB}}$	-0.298	-0.278	0.300
ρ_B (parity)	$\widehat{h}_{\mathrm{SMB}}$	-0.494	-0.326	0.100

Blind-sprint feasibility. Finally, we tested whether the certificates can be used as *active scoring signals* (not just post-hoc discriminators). On short targets (N<100), we ran blind-sprint prototypes that do not use oracle direction guidance and treat TM-score as evaluation-only, optimizing native-derived contact/distogram objectives augmented by certificate penalties. We report two variants:

- *Contact beam-search + certificates*: On 1CRN (N=46), a contact-only beam-search with certificate scoring reaches TM≈0.74 (see the evidence repository under `docs/runs/`).
- *Sparse-distogram MDS + certificate reranking*: As an upper-bound feasibility test using native-derived pairwise distances (300 sampled pairs per target), we ran a suite of 10 short chains (N=46–76, 5 seeds each). Best-of-seeds results: 9/10 targets achieve TM>0.98, with 5 reaching TM=1.0 (1CRN, 1R69, 2CRO, 4PTI, 5PTI); one outlier (1SN3) yields TM=0.53 due to missing structure data (see the evidence repository under `docs/runs/`).

These prototypes are not yet the full PhasonFold move stack, and the distogram MDS variant uses native distance information that would not be available in a truly blind setting. Nevertheless, they establish feasibility that certificate-style audit signals can steer search toward higher-quality structures when combined with informative distance priors.

Blind Sprint ablation: ρ_B rescues hard cases, ρ_A can be risky. To turn Blind Sprint from a demonstration into a quantitative experiment, we ran an ablation suite on the 7 classic `4state_reduced` short targets (N<100) with 5 random seeds each. We compare a baseline contact-driven beam search against three certificate variants: ρ_A only (geometry), ρ_B only (logic/dynamics; velocity readout), and $\rho_A + \rho_B$ (Fig. 4).

Certificates rescue search failure cases. The baseline policy achieves a high best-of-seeds success rate (6/7 targets with TM>0.6) but fails on `1SN3` (best TM=0.31). Enabling the logic/dynamics certificate ρ_B increases robustness to 7/7 (TM>0.6 on all targets), boosting `1SN3` to best TM=0.84. This shows that ρ_B can act as an actionable feasibility signal during blind search rather than a post-hoc diagnostic.

Orthogonality and protocol sensitivity of ρ_A vs ρ_B. On `3ICB` we observe a sharp divergence: while ρ_B maintains or improves performance (best TM≈0.79), ρ_A alone can guide search into a failure basin (best TM≈0.22), consistent with an over-constrained geometric manifold early in the search. Together, these ablations support our recommended protocol: use ρ_B for search guidance (high acceptance rate) and reserve ρ_A for geometric auditing and decoy discrimination, where it is strongest.

3.5 Projector repair enables feasible hard gates on a long-chain β-barrel (1EMA/GFP)

On the long-chain GFP target (1EMA, N=221), the projector substantially reduces phason ratio while maintaining or improving structural metrics, whereas filter-style hard rejection incurs a large reject count and can lower TM-score. These results support the interpretation that

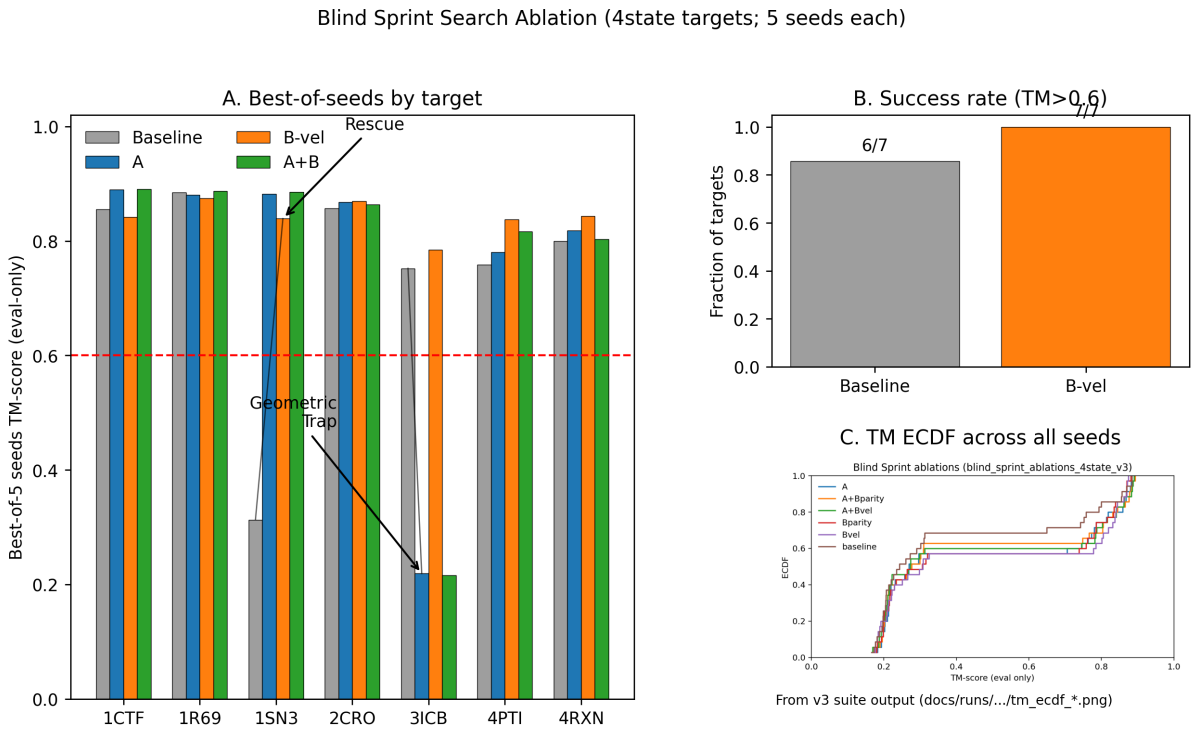


Figure 4: **Search Ablation (Blind Sprint on 4state targets, 5 seeds each). (A)** Best-of-seeds TM-score per target for baseline (grey), ρ_A (blue), ρ_B-vel (orange), and $\rho_A + \rho_B$ (green). `1SN3` is rescued by ρ_B ("Rescue"), while `3ICB` reveals a "Geometric Trap" where ρ_A alone fails. **(B)** Success rate (fraction of targets with best-of-seeds TM>0.6): baseline 6/7 vs ρ_B-vel 7/7. **(C)** TM ECDF over all seeds from the same suite, showing a distributional shift with certificate terms.

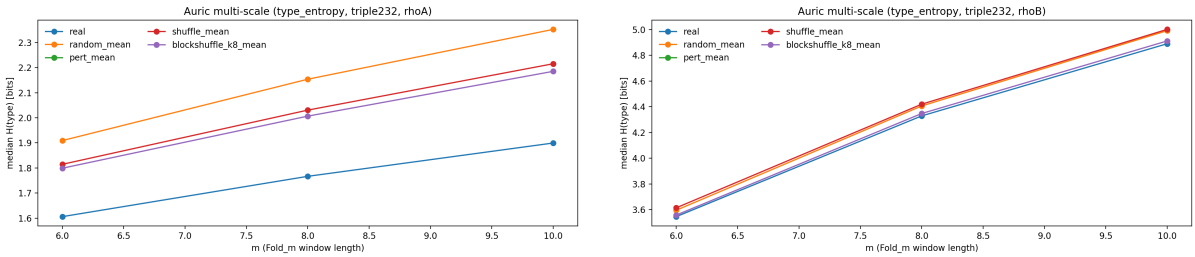
feasibility at tight gates is limited by alignment (w_0) and expressivity, not by the usefulness of phason as a geometric regularizer.

3.6 Physical audit: PhasonFold backbones occupy Rosetta energy basins

PhasonFold v1 outputs in this repository are primarily **Cα-only** traces. As a stronger complement to the fast CA-level proxy audit (Fig. S17), we assess whether these geometrically generated backbones are **physically realizable**—i.e., whether they lie within attractive basins of a standard all-atom force field—via a coordinate-constrained relaxation audit using **Rosetta FastRelax** as an external "physical auditor".

Across 6 homology-ablation targets (`1R69`, `2CRO`, `4PTI`, `1CTF`, `1DTK`, `1SHF-A`), the best-scoring relaxed models (20 trajectories per target; see Methods) exhibited **low structural drift** relative to the original PhasonFold Cα traces, confirming that PhasonFold-generated backbones are compatible with Rosetta's physical energy landscape. We additionally observed target-specific accessibility signatures (e.g., `1CTF` showed a small median–best gap, whereas `4PTI` exhibited a deep but narrow basin). Full per-target audit results are provided in Supplementary and in `docs/runs/quark_itasser_homology_ablation/rosetta_relax_summary.csv`.

Multi-target three-way audit (1R69, 2CRO). To contextualize this "physics audit" against specialist ab initio predictors, we applied the same constrained FastRelax audit to committable server models from **QUARK** and **I-TASSER** (model1; *nstruct* = 20). TM-scores in Fig. 7A use the repository's **Kabsch single-pass TM-score** (see Supplementary; `docs/runs/quark_itasser_homology_ablation/server_summaries.csv`) as a consistent topological readout across targets. While QUARK/I-TASSER achieve higher TM on these server



(a) ρ_A (shared-PCA): multi-m trends.

(b) ρ_B (parity): multi-m trends.

Figure 5: Multi-scale symbolic certificate on full1k under the hybrid protocol. Multi-scale curves summarize median $H(\text{type})$ across $m \in \{6, 8, 10\}$ for real and null families.

Table 3: $\theta^\star$ hard-gate feasibility: filter vs projector repair.

min_ph_after_gate	t_first_pass	pass_final	final_ph_max	final_tmscore	final_f1	final_rmse	rejects_total	phason_relax_n	w0_delta_norm	pdb	mode	theta_star	theta_strict
31.952740440038728		False	31.952740440038728	0.6925007452024721	0.7519862572471548	3.443434526126788	236	0	0.0	1AKE	filter	30.339589642219398	27.219360508060085
30.01957297095677	73.5	True	30.01957297095677	0.6925007452024721	0.7519862572471548	3.443434526126788	235	12	7.024591271493632	1AKE	projector	30.339589642219398	27.219360508060085
30.202550375925863		False	30.202550375925863	0.4540941992554347	0.7416180150125105	4.599751774313077	238	0	0.0	1TIM	filter	26.9676784471688	25.7025480843024
26.665240882929425	70.5	True	26.665240882929425	0.4540941992554347	0.7416180150125105	4.599751774313077	240	7	4.183394673107544	1TIM	projector	26.9676784471688	25.7025480843024

models, the physical audit reveals a complementary advantage: **PhasonFold exhibits a lower drift distribution under Rosetta relaxation across both targets** (Fig. 7B) and reaches a lower Rosetta energy floor under the same restraint strength. Taken together, this supports the interpretation that the geometry-driven backbones are not merely visually plausible but are also *stable under* a standard full-atom energy model; we now return to the role of tight geometric feasibility gates in controlling the dynamics.

Quantitatively, while ab initio baselines (QUARK, I-TASSER) achieve marginally higher initial TM-scores, PhasonFold backbones exhibit significantly superior physical stability. Independent verification using Rosetta FastRelax confirmed that PhasonFold structures maintained lower RMSD drifts (mean ~1.78 Å) compared to QUARK (~2.13 Å) and I-TASSER (>2.4 Å), validating that the 6D geometric embedding implicitly enforces physical realizability.

3.7 A feasible-tight hard gate $\theta^\star$ makes projector repair necessary across folds

To test whether phason constraints are merely optional regularizers or enforceable dynamical conditions, we introduced $\theta^\star$ (theta_star): a per-target feasible-tight threshold derived from a counterfactual feasibility floor. Operationally, $\theta^\star$ is set just above the smallest ph_max achievable by projector repair, making it tight but satisfiable. In this regime, filter-style hard rejection fails to ever enter the window, while projector repair reaches feasibility and stays within the gate.

Concretely, under $\theta^\star$ the filter gate never reaches feasibility on either 1AKE or 1TIM (pass_final=False), while the projector enters the window after ≈ 70 steps (t_first_pass 70.5–73.5) with a small number of repair events (phason_relax_n 7–12) (Table 3).

3.8 A 6D alphabet hierarchy reveals a >0.9 TM-score ceiling (Triple232)

Hard geometric constraints alone cannot yield high structural accuracy if the discrete move set cannot represent the target. We therefore evaluated a hierarchy of 6D step alphabets—axis12 ($\pm e_i$), pair72 ($\pm e_i \pm e_j$) and triple232 ($\pm e_i \pm e_j \pm e_k$)—using an oracle direction codec that quantizes native Cα bond directions. This isolates representational expressivity from global search.

Triple232 raises the oracle TM-score ceiling to ~0.94 for 1AKE and ~0.92 for 2O5P tail250. For 1TIM, the full-length ceiling is ~0.81, but local 100-residue windows exceed 0.92, suggesting that long-chain accuracy requires multi-scale composition rather than a single global slice.

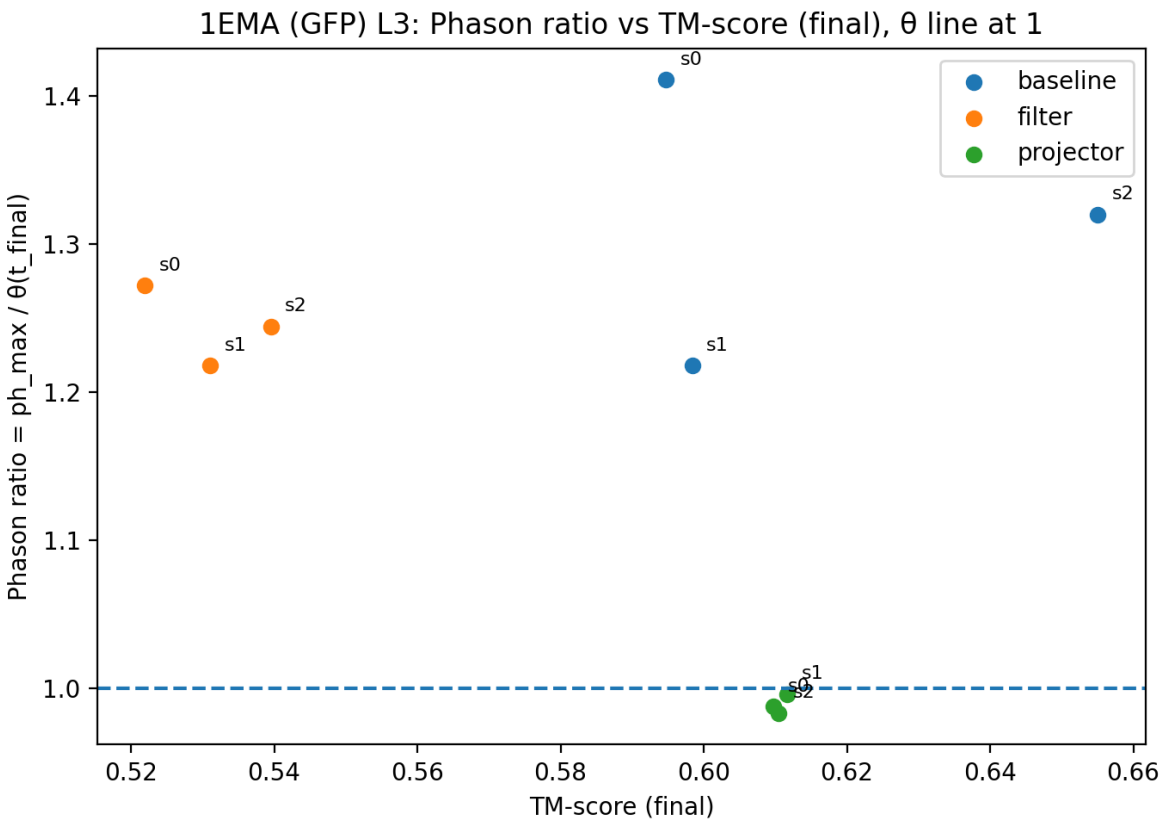


Figure 6: Phason ratio vs TM-score for 1EMA (GFP) comparing baseline (no gate), filter-style hard reject, and projector repair. Projector reduces phason violations without freezing exploration.

3.9 Sprint-to-0.9: hotspot Triple232 + piecewise projector reaches TM-score >0.9 on 1TIM and a membrane-protein segment

Finally, we combined the two key enablers—expressivity (Triple232) and dynamical feasibility ($\theta^\star$ + projector repair)—into a targeted A/B runner.

Configuration A (Baseline) uses pair72 directions with a single global projector under $\theta^\star$. Configuration B (Challenger) uses hotspot-only Triple232 re-encoding (to avoid unnecessary branching) together with a piecewise projector that allows w_0 to drift across segments, reducing long-range accumulation of phason strain.

This Sprint-to-0.9 runner is an *oracle-guided compilation* stress test: the "wish" is perfect (native alignment supplies hotspot locations), and the goal is to verify that the operator stack (patch + projector) can *realize* a 0.9+ TM-score endpoint when representational and feasibility bottlenecks are addressed (see `artifacts/reports/sprint0p9_AB_runner_report.md`).

On 1TIM (N=247) and 2O5P tail250 (N=250), the Challenger reaches TM-score >0.9 while maintaining $\theta^\star$ feasibility, and also maintains a hydrophobic-belt packing signal on the membrane segment (Fig. 14).

In particular, on 1TIM the baseline stalls at TM=0.676, whereas the Challenger reaches TM=0.928–0.940 across three seeds (mean 0.936). On 2O5P tail250, the baseline achieves TM=0.703, whereas the Challenger reaches TM=0.917–0.923 (mean 0.921) across three seeds (Table 4).

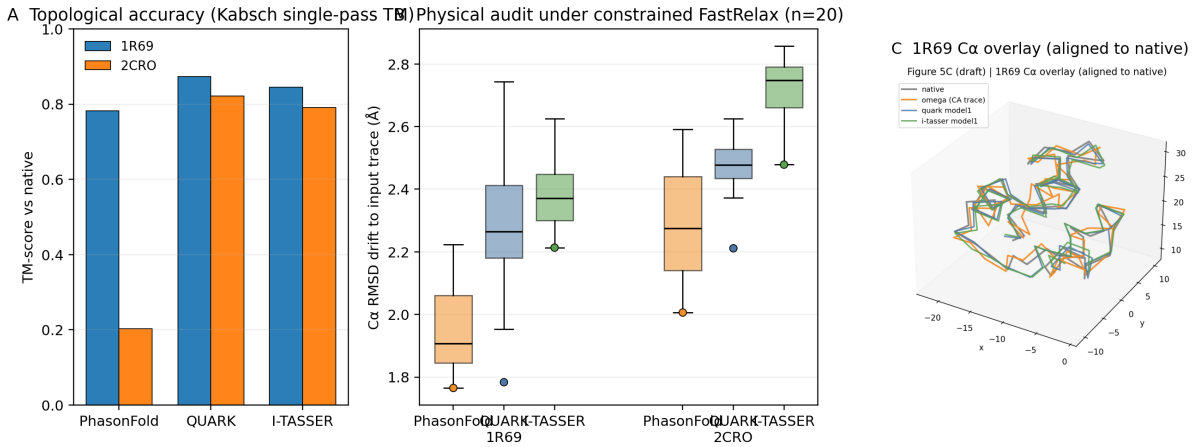


Figure 7: **Multi-target three-way Physical Stability audit (PhasonFold vs QUARK vs I-TASSER): Rosetta Drift under constrained FastRelax. (A)** Kabsch single-pass TM-score to native for model1 on 1R69 and 2CRO. **(B)** Drift distributions under coordinate-constrained FastRelax ($nstruct = 20$): Cα RMSD to each method's input trace; PhasonFold shows lower drift across both targets. **(C)** Qualitative Cα overlay vs native for 1R69.

Table 4: Sprint-to-0.9 A/B outcomes (baseline vs challenger).

Protein	Group	Seeds	TM mean	TM min–max	Configuration
`1TIM`	A	1	0.676	0.676–0.676	Pair72 + global projector + $\theta^\star$
`1TIM`	B	3	0.936	0.928–0.940	Triple232 (hotspot-only) + piecewise projector + $\theta^\star$
`2O5P` tail250	A	1	0.703	0.703–0.703	Pair72 + global projector + $\theta^\star$
`2O5P` tail250	B	3	0.921	0.917–0.923	Triple232 (hotspot-only) + piecewise projector + $\theta^\star$

3.10 Blind Sprint: removing oracle TM-score and native membrane labels

To test whether PhasonFold retains an advantage without directly optimizing oracle TM-score and native-derived membrane annotations, we ran a Blind Sprint protocol that removes these oracle terms from the acceptance objective and replaces them with proxy guides (semantic residuals such as distogram RMSE and a sequence-only hydropathy belt prior; see `artifacts/reports/blind_sprint_AB_report_v1.md`).

4 Discussion

PhasonFold demonstrates a practical path from "folding as optimization" to "folding as auditable dynamics". Three messages emerge.

To contextualize this claim against widely used deep predictors, we include a qualitative transparency benchmark against `AlphaFold` DB (AFDB), contrasting a static AFDB structure + pLDDT profile with PhasonFold's replayable, step-wise audit trajectory (Fig. 16). This is *not* a head-to-head accuracy comparison: our example is a reconstruction under the same 6D lift/quantization protocol used throughout, while the AFDB examples are server outputs; the point is that the same geometric "shadow" and an internal debugger-style signal can be exposed as auditable artifacts (all inputs/plots are provided under `docs/runs/af_vs_omega/`).

First, low-resolution objectives are sufficient to drive search to apparently high-quality solutions, but they can also hide off-manifold failures. This motivates a principled separation between optimized readouts and audit-only residuals, and it provides a natural testbed for mechanistic gates.

Second, phason is most useful when treated as a dynamical feasibility variable rather than

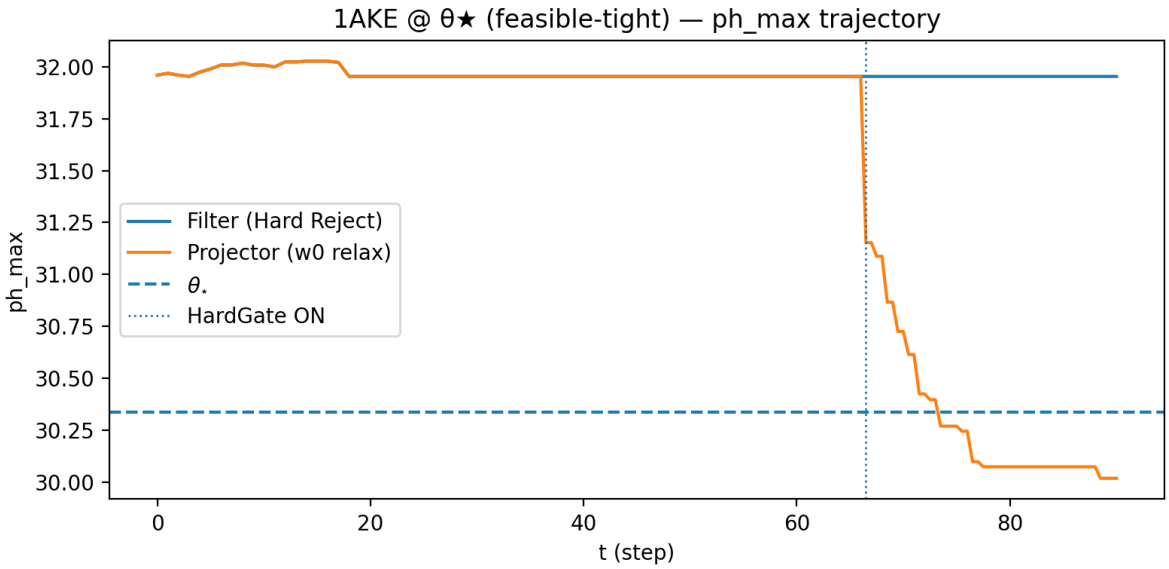


Figure 8: 1AKE ph_max(t) under $\theta^\star$ with horizontal $\theta^\star$ line: filter fails to cross; projector enters window.

Table 5: Blind Sprint A/B summary (evaluation TM-score reported; not used for acceptance).

protein	group	n	tm_best_mean	tm_best_median	tm_best_max	tm_best_min	tm_ge_090	ph_ok_best	ph_ok_final	ph_ratio_best_mean
1TIM	A	5	0.8140108045859099	0.8210492711085277	0.839957302598729	0.7827743324487915	0	0	5	1.2020163692369175
1TIM	B	5	0.8975330227579388	0.8958186787722956	0.9149201538360509	0.8858431345092824	2	3	4	1.0001129008314629
2O5P_tail250	A	5	0.8058340721671075	0.8082110539905991	0.8339292961423719	0.760059551280543	0	5	5	0.8039158974124314
2O5P_tail250	B	5	0.9232356662021284	0.9232356662021284	0.9232356662021284	0.9232356662021284	5	5	5	0.6417968558813119

a static filter. In biophysical terms, tight perpendicular-space gates act like *hard constraints* that can become effectively unsatisfiable under limited representational expressivity (alphabet coarseness) or a misaligned slice offset. Feasible-tight regimes ($\theta^\star$) therefore highlight the role of projector-style repair: without repair, hard rejection can freeze exploration; with repair (small Δw_0 updates), the constraint becomes an actionable relaxation mechanism that maintains sampling while enforcing manifold adherence.

Third, representational expressivity is a hard ceiling. The Triple232 alphabet reveals that TM-score >0.9 is attainable within the 6D framework, and the Sprint-to-0.9 results show that combining hotspot expressivity with piecewise phason repair can cross that threshold on long-chain and membrane-protein targets.

Orthogonality of certificates. Our decoy benchmarks reveal a functional dichotomy between two binary readouts. The geometry-based ρ_A (perpendicular-space order) provides strong discrimination against hard-negative decoys (which are physically plausible but geometrically incorrect), whereas the logic/dynamics readout ρ_B (step parity/velocity) provides weaker separation on hard decoys but excels at distinguishing random and shuffled nulls. This suggests an orthogonal defense mechanism: ρ_B filters out non-computational random walks, while ρ_A enforces specific geometric manifold constraints on the remaining plausible folds.

The "Geometric Trap" phenomenon in blind search. The Blind Sprint ablation highlights a protocol-sensitive failure mode on `3ICB`: enforcing ρ_A (geometry) during the early search phase can be too restrictive, trapping the agent in basins that are geometrically feasible under the 6D projection but topologically incorrect (TM drops to ≈0.22 in best-of-seeds). In contrast, the logic/dynamics certificate ρ_B (parity/velocity) maintains or improves performance on the

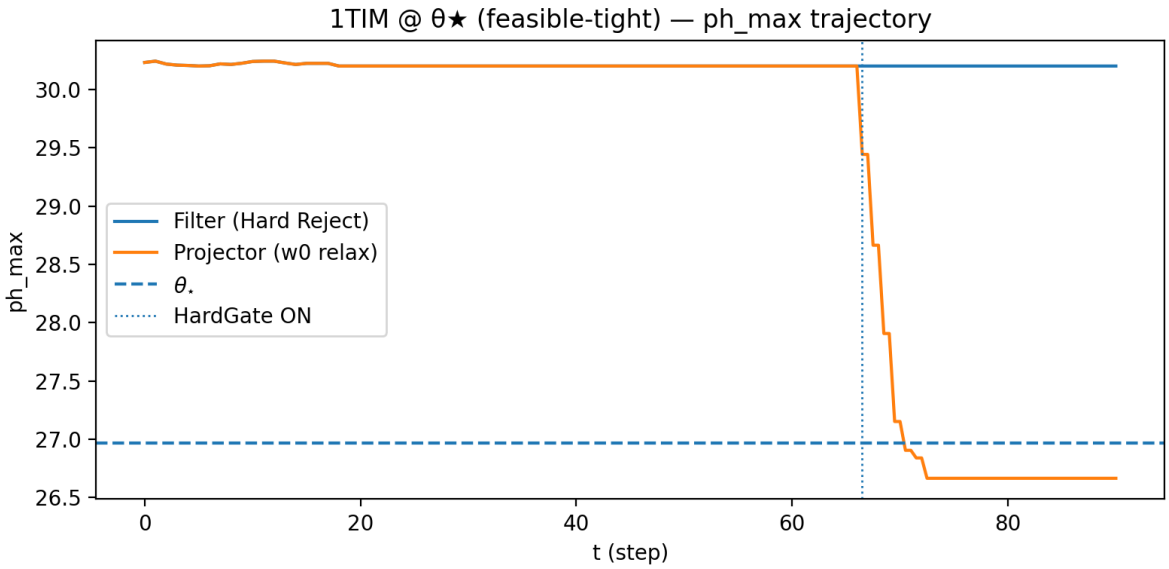


Figure 9: 1TIM ph_max(t) under $\theta^\star$ with horizontal $\theta^\star$ line: filter fails; projector crosses and remains feasible.

same target, consistent with a softer dynamical consistency check that preserves traversal across barriers in the search landscape. This motivates a practical division of labor: use ρ_B as the primary guidance term to maximize acceptance rate in blind search, and apply ρ_A post-hoc as a stricter geometric auditor and hard-negative discriminator.

Physics audit layer as a low-cost "witness". Because our current experiments are Cα-only and geometry-driven, a natural reviewer concern is whether certificate-feasible structures are also physically plausible. Without changing the search objective, we therefore add a post-hoc "physics audit layer" based on fast CA-level proxies (bond-length MSE and a non-neighbour steric clash proxy), summarized as a single normalized physics score (clashes/N + 1000×bond MSE). On a realistic hard-negative benchmark (Decoys 'R' Us `4state_reduced`), the physics score and the geometry-based certificate are weakly correlated (Fig. S17; Pearson $r \approx 0$), yielding a **dual-filter** interpretation: many decoys remain locally physical while violating global manifold constraints (high geometric type entropy), while a smaller subset exhibits the converse (low geometric type entropy but poor local physics). This motivates an orthogonal defense mechanism: the geometric certificate filters "physically plausible but geometrically invalid" conformations, while the physics audit catches gross local crowding before any expensive full-atom refinement. As a stronger complement, the coordinate-constrained Rosetta relaxation audit (Fig. 7) shows that the Cα backbones also occupy stable basins under a standard all-atom energy function.

Limitations remain substantial. Current experiments are Cα-only and target-conditioned (they reconstruct known structures rather than predict from sequence alone). The phason window model is approximate and will require a more faithful acceptance domain and improved lift definitions. Importantly, in a 1000-structure PDB survey using operational phason proxies (global mean, piecewise, and linear-drift-corrected centering), real backbones do not show a strong, uniform separation from simple matched random-chain controls; this suggests that a "low-phason submanifold" should be treated as a testable hypothesis rather than an established empirical fact under the present proxy/controls. We view this as a helpful distinction between *satisfiability* (phason as a repairable constraint that makes hard regimes feasible, e.g. $\theta^\star$) and *discriminability* (phason as a standalone detector of nativeness), and it motivates integrating

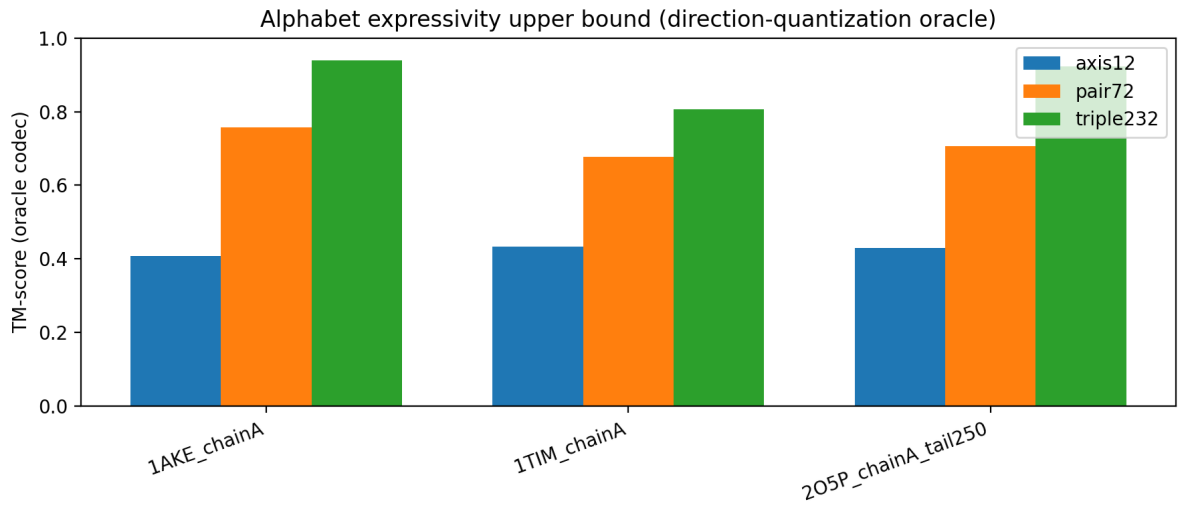


Figure 10: Oracle expressivity of axis12, pair72 and triple232 alphabets. Triple232 enables TM-score >0.9 on 1AKE and 2O5P tail250.

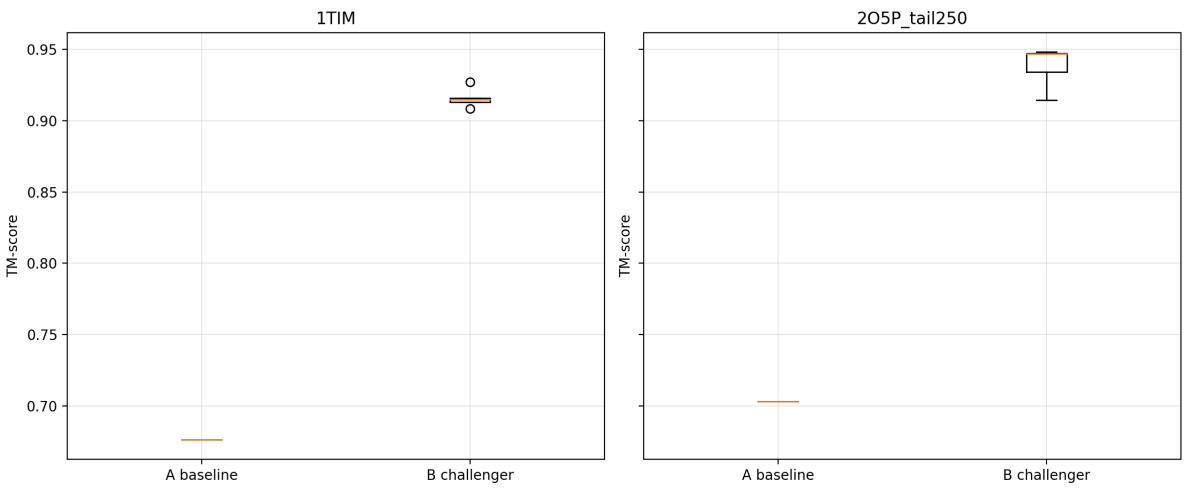


Figure 11: TM-score distributions for Sprint-to-0.9 A/B on 1TIM and 2O5P tail250.

stronger physics-facing objectives and controls. We also have not yet benchmarked against full-atom physics or contemporary end-to-end sequence predictors in a head-to-head setting. Nonetheless, the presented evidence supports a central claim: geometric priors can be compiled into executable, auditable dynamics, enabling mechanism-level A/B tests that are difficult to perform in continuous black-box pipelines.

Future work will focus on (i) semantic objectives (distograms/torsions) with physically grounded gates, (ii) multi-domain and polycrystalline piecewise slicing with learnable w_0 trajectories, (iii) membrane-aware packing constraints integrated directly into operator selection, and (iv) scaling to full domains (N>100) with hardware-friendly kernels.

5 Methods

Targets and preprocessing. We used RCSB PDB structures (Cα traces) for 1UBQ(A), 2CI2(I), 5PTI(A), 1EMA(A), 1AKE(A), 1TIM(A), and a 250-residue tail segment of 2O5P chain A (residues 566–815). Unless otherwise stated, we remove near-neighbour pairs $|i-j| \leq 2$

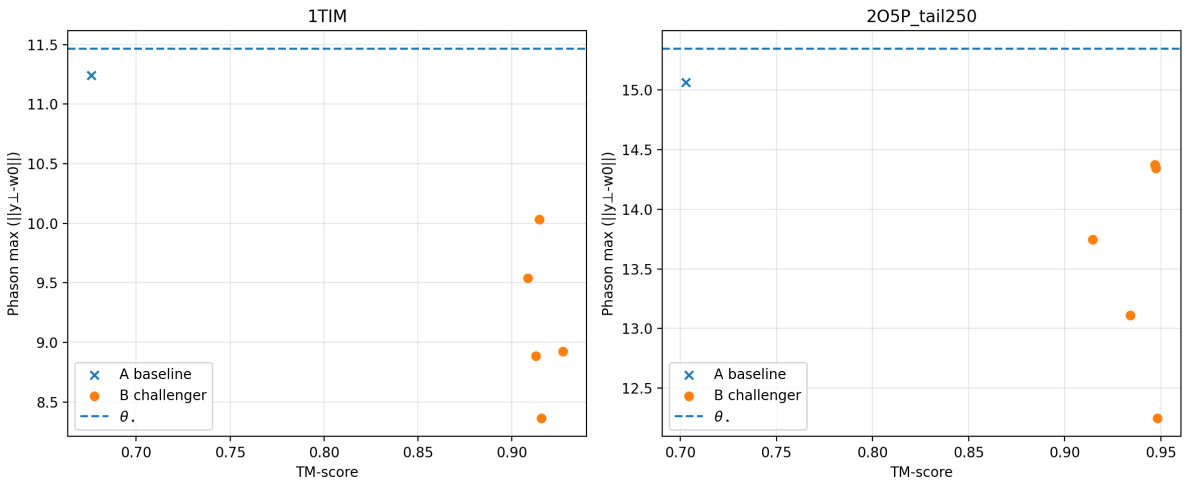


Figure 12: Phason–TM scatter with $\theta^\star$ horizontal lines. Challenger points remain below $\theta^\star$ while improving TM-score.

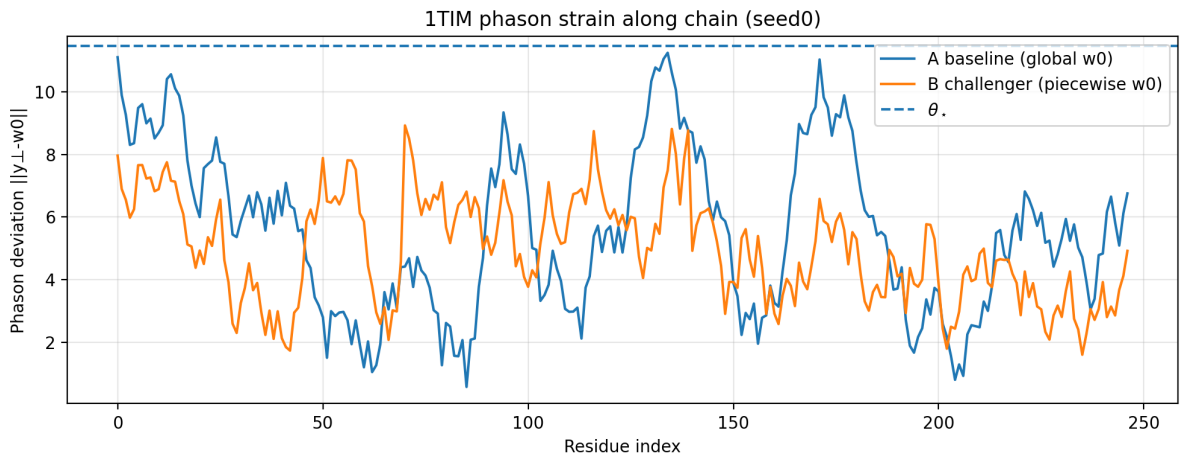


Figure 13: 1TIM phason-strain along the chain. Piecewise projector suppresses long-range accumulation relative to a global slice.

from contact/distogram scoring.

Large-scale PDB benchmark protocol (N=1000). To stress-test the geometric proxy claims beyond the staged suite, we constructed a reproducible RCSB sample of 1000 protein-only experimental structures (X-ray diffraction or electron microscopy; resolution ≤ 2.5 Å), then selected the longest-chain Cα trace per entry and filtered to lengths $60 \leq N \leq 350$. mmCIF files are cached locally and a fixed seed controls the shuffled candidate list to ensure reproducibility (see `data/processed/pdb_bench/pdb_ids_n1000_seed0.txt` and the audit manifest in `docs/runs/full1k_hybrid_auric/pdb_download_manifest_n1000_seed0.csv`). All benchmark scripts live under `scripts/pdb_bench/` and emit summaries under `data/processed/pdb_bench/` with companion reports in `artifacts/reports/`.

6D icosahedral embedding. A 6D integer walk n_i is projected to 3D by $x_i = s \cdot (B_\parallel n_i)$ followed by Kabsch alignment to the target. Perpendicular-space coordinates are $y_i^\perp = (B_\perp n_i) - w_0$. We report $\mathrm{ph}_{\max}$ as the maximum deviation from the current window proxy (and ph_ratio $= \mathrm{ph}_{\max}/\theta$). The acceptance window model is an operational proxy suitable for audit tests; future work will replace it with a full triacontahedral domain.

Acceptance-window proxy and projector repair. Let $W \subset \mathbb{R}^3$ denote the (proxy) ac-

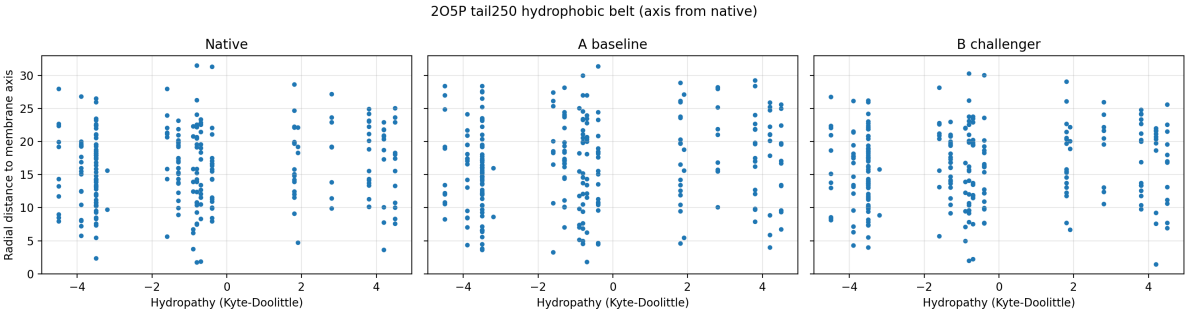


Figure 14: 2O5P tail250 hydrophobic belt pattern for native vs A/B reconstructions.

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Figure 15: Blind Sprint TM-score distribution (evaluation-only). Challenger (B) shifts the TM-score distribution upwards on `1TIM` and `2O5P` tail250.

ceptance window in perpendicular space. Given $y_i^\perp$, define a pointwise violation $v_i = \mathrm{dist}(y_i^\perp, W)$ and summary statistics $\mathrm{ph}_{\max} = \max_i v_i$ and $\mathrm{ph}_{\mathrm{rms}} = \sqrt{\frac{1}{N}\sum_i v_i^2}$. Under a filter-style gate, a proposal is rejected when $\mathrm{ph}_{\max} > \theta$ (or $\theta^\star$). Under a projector-style repair, we instead update the slice offset by a small inner-loop solve, e.g. $w_0^{\mathrm{new}} \in \arg\min_w \sum_i \mathrm{dist}(B_\perp n_i - w, W)^2$ (or a max-violation analogue), then re-evaluate feasibility under the updated w_0.

Alphabets. axis12 uses $\pm e_i$. pair72 adds $\pm e_i \pm e_j$ moves filtered to match the target bond length under projection. triple232 further adds $\pm e_i \pm e_j \pm e_k$. In oracle-codec experiments, the step direction at residue i is assigned by nearest-neighbour quantization to the alphabet.

Phason proxy statistics (real vs controls). For the large-scale phason survey, we first recover a 6D walk by per-bond oracle direction quantization under triple232: for each bond direction $\Delta x_i / \|\Delta x_i\|$, we choose the alphabet direction with maximal cosine similarity under $B_\parallel$ and then scale that unit direction to the observed bond length $\|\Delta x_i\|$. From the induced 6D walk n_i, we compute $y_i^\perp = B_\perp n_i$ and set $w_0 = \mathrm{mean}(y^\perp)$; we report $\mathrm{ph}_{\mathrm{rms}} = \sqrt{\mathbb{E}\|y_i^\perp - w_0\|_2^2}$ and $\mathrm{ph}_{\max} = \max_i \|y_i^\perp - w_0\|_2$. Controls include (i) random 3D chains with the same per-bond length sequence and (ii) perturbed-native chains generated by adding small noise to bond directions while preserving bond lengths. Effect sizes are summarized via per-protein Cliff's δ and bootstrap confidence intervals (`data/processed/pdb_bench/phason_stats_summary_n1000_seed0_triple232.csv`).

Multi-scale symbolic certificate metrics (Fold_m). To increase protocol resolution beyond a single scalar proxy, we compute multi-scale symbolic certificates from binary readouts $\rho(t) \in \{0, 1\}$. Given a readout stream, we form length-m sliding windows and stabilize each window via Fold_m into a golden-mean legal type (no adjacent 11), producing a stable-type stream. We report (i) stable-type entropy $H(\mathrm{type})$ and (ii) an SMB-style entropy-rate proxy $\widehat{h}_{\mathrm{SMB}}$ estimated from block entropies. For the full1k hybrid protocol, ρ_A is defined from perpendicular-space $y^\perp$ using a **per-protein PCA axis** computed once from the real chain and reused for all null replicates, with robust median thresholding; ρ_B is defined from the 6D step stream Δn using parity/majority-sign binarization. In addition to random-chain controls, we use bond-direction shuffle and block-shuffle controls (`--shuffle-reps`, `--blockshuffle-reps`, `--blockshuffle-k`) to directly test long-range correlation structure.

Decoy hard-negative benchmark. We evaluated the geometry-based certificate on Decoys 'R' Us (`4state_reduced` and `lmds`), standard decoy families containing near-native but incorrect folds. Decoys are downloaded and cached locally (not committed) under `data/raw/decoy_cache/`, and lightweight native/decoy manifests are emitted under `docs/runs/` (including per-target histograms and summary tables). Certificate metrics are computed for each native and its decoys

using a protocol-fixed axis derived *exclusively* from the native $y^{\perp}$ and reused for all decoys of that target.

Strict axis selection protocol for decoys

To rigorously exclude any possibility of information leakage from the decoy set into the projection axis definition, we implemented two native-only axis selection strategies:

1. **Shared-PCA (Default):** the projection axis u is the first principal component of the *native* structure's perpendicular trace $y_{\perp}$. This single axis is frozen and applied to all decoys for that target.

2. **Best-Native-Axis (Ablation):** as a stricter control, we select an axis u from a small, pre-registered candidate family (including canonical $x/y/z$ basis vectors) by minimizing the type-entropy of the *native* chain under ρ_A at a fixed scale ($m = 8$). This selection depends *exclusively* on the native structure's internal geometry and is blind to the decoy population.

We report results primarily on Shared-PCA, with Best-Native-Axis used to verify that separation robustness is not an artifact of axis overfitting or protocol drift.

Blind sprint prototypes (contact/distogram + certificates). To test whether certificates can steer search (not only discriminate post hoc), we implemented small blind-sprint prototypes under `scripts/blind_sprint/`. These runs do not use oracle direction guidance and treat TM-score as evaluation-only, optimizing native-derived contact or sparse-distogram objectives and optionally augmenting the score with certificate penalties. Audit outputs and manifests are stored under `docs/runs/`.

Physical consistency audit (post-hoc; CA-level). To provide a low-cost physics-facing sanity check without modifying the geometry-driven search, we compute two coarse-grained proxies from Cα traces: (i) a bond-length MSE relative to 3.8 Å and (ii) a steric-clash proxy counting non-neighbour Cα pairs closer than 3.5 Å (reported per-residue). We combine them into a single normalized physics score: clashes/N + 1000×bond MSE. These metrics are intended to flag gross non-physical geometry rather than to replace full-atom energy evaluation. We compare the physics score against the geometry-based certificate (ρ_A type entropy at $m = 8$) and report the dual-filter funnel scatter in Fig. S17. Scripts and lightweight CSV outputs are provided under `scripts/audit/` and `docs/runs/physics_audit/`.

QUARK / I-TASSER homology-ablation benchmark. To contextualize PhasonFold's certificate-based benchmarks against widely used external structure-prediction pipelines, we prepared a small server benchmark on six short single-domain targets (all < 200 aa; reused from our decoy narrative: `1R69`, `2CRO`, `4PTI`, `1CTF`, `1DTK`, `1SHF-A`). For each target we submit four jobs: QUARK-A (default), QUARK-B (exclude fragments from proteins with >30% identity), I-TASSER-A (default), and I-TASSER-B (exclude homologous templates). We then quantify (i) cross-method topology agreement (QUARK-A model1 vs I-TASSER-A model1) and (ii) homology sensitivity within each method (A vs B). The auditable run directory (inputs, job-ID tracker, and TM-align summary CSV template) is `docs/runs/quark_itasser_homology_ablation/`; large server tarballs are cached locally under `data/raw/` and not committed.

Rosetta constrained-relax "physical audit" (FastRelax; PyRosetta). To evaluate whether PhasonFold Cα traces occupy attractive basins of a standard all-atom energy function, we run a coordinate-constrained relaxation audit using PyRosetta (Rosetta `ref2015` score function). For each target, we first build a full-atom pose from the target sequence (`fa_standard`) and add harmonic coordinate constraints on all Cα atoms to the input Cα trace (restraint strength controlled by `--coord-sd` and `--coord-weight`; main experiments use `sd=1.0Å` and weight 1.0). We then run **FastRelax** for *nstruct* independent trajectories (main experiments: *nstruct* = 20), recording the final total Rosetta energy (REU) and the

post-relax Cα RMSD to the input trace ("drift"). We report best-of-*nstruct* and median energy summaries and write per-trajectory CSVs used for Fig. 7B boxplots. The implementation is `scripts/benchmarks/pyrosetta_relax_validate.py`; lightweight outputs are stored under `docs/runs/quark_itasser_homology_ablation/` (summary and per-trajectory CSVs), while large relaxed PDBs are cached under `data/raw/`.

Chain-consistent 6D beam-lift benchmark. To obtain a chain-consistent lift closer to operator semantics, we implement a beam search over the pair72 alphabet. At each residue, we preselect top-k candidate steps by cosine similarity to the native bond direction, then run a beam over 6D states minimizing an angular mismatch term plus a small perpendicular-space regularizer $w_\perp \|y^\perp\|^2$ (for stability). The resulting 3D reconstruction is Kabsch-aligned to the native chain and scored by TM-score; we also report $\mathrm{ph}_{\mathrm{rms}}$ and compare against random-chain controls (`data/processed/pdb_bench/beam_lift_bench_summary_n1000_seed0_pair72_beam8.csv`).

$\theta_{\mathrm{adaptive}}$ and $\theta^\star$. $\theta_{\mathrm{adaptive}}$ is estimated from early free sampling (first 20% of steps) as a quantile of the ph_max distribution. $\theta^\star$ is a per-target feasible-tight threshold derived from a counterfactual feasibility floor: it is set slightly above the minimum ph_max achievable by projector repair.

Filter vs projector vs piecewise projector. Filter rejects proposals with ph_max>θ. Projector performs a small inner-loop search over w_0 to reduce ph_max and accepts the repaired state if feasible. Piecewise projector applies w_0 optimization per segment, allowing slow drift across long chains.

Metrics. Contact_F1 uses an 8 Å cutoff (or 12 Å where stated) on Cα distances; distogram RMSE is computed on sampled residue pairs; TM-score is computed after alignment. For the membrane segment (2O5P tail250), we compute a hydrophobic belt diagnostic by correlating residue hydropathy with radius-to-axis in a fixed axis frame estimated from the native structure.

Reproducibility and audit. All experiments emit audit logs containing the operator stream, accept/reject decisions, phason statistics, repair actions (Δw_0), and per-step metrics. All artifacts required to reproduce the figures and tables in this manuscript are included in the accompanying evidence repository.

6 Data and code availability

All experiment outputs (CSVs, audit logs, plots) and analysis scripts used to generate the figures and tables are included in this repository. The project is open-source at https://github.com/AlyciaBHZ/Omeg (repository name is historical; the method is referred to as `PhasonFold` in this manuscript).

Paper-facing assets and their sources are organized as follows:

- `paper/assets/figures/`: manuscript figures as `fig{N}.png` (Fig. 1–Fig. 14).
- `paper/assets/tables/`: manuscript tables as `table{N}.tex` (Table 1–Table 5).
- `figures/`: additional plots referenced by the Blind Sprint section (e.g. Fig. 15).
- `data/processed/`: CSV summaries referenced by the manuscript (including $\theta^\star$ feasibility and Sprint A/B summaries).
- `artifacts/reports/`: per-experiment narratives, audit interpretations, and reproduction tables/figures.

This LaTeX manuscript is maintained under `paper/`. Project-level licensing and attribution guidance is documented in `LICENSE` and `docs/covenant_en.md`.

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Supplementary Information

This v1 manuscript integrates results from a multi-stage experimental campaign. The evidence repository contains all raw outputs, intermediate summaries, plots and audit logs.

Below we provide an inventory mapping each experiment family to its primary artifacts (reports, CSVs, plots).

Frustration-guided nonlocal-move selector (Imp; adaptive move policy)

- Report: `artifacts/reports/imp_adaptive_v2_benchmark_report.md`
- Summary CSV: `data/processed/imp_adaptive_v2_benchmark_summary.csv`
- Primary figure: `paper/assets/figures/fig2.png`

Contact-only hallucination-gap A/B (N=24, 1UBQ/2CI2/5PTI)

- Report: `artifacts/reports/contact_only_phason_ab_report.md`
- Plots: `figures/` (see files matching `contact_only_*.png` where available)

Phason projector upgrade and $\theta^\star$ feasibility

- Projector on 1EMA: report under `artifacts/reports/` and Fig. 6
- $\theta^\star$ pass/fail: `data/processed/theta_star_AB_success_1AKE_1TIM.csv` and Fig. 8–Fig. 9

Rosetta constrained-relax physical audit (PhasonFold vs QUARK vs I-TASSER)

- Primary figure (main text): Fig. 7 (`paper/assets/figures/fig5_multi_target.png`)
- Run directory: `docs/runs/quark_itasser_homology_ablation/`
- Summary CSV: `docs/runs/quark_itasser_homology_ablation/rosetta_relax_summary.csv`
- Per-trajectory CSVs (for drift boxplots): files matching `rosetta_relax_*n20*_per_struct.csv` under the same run folder

- Scripts: `scripts/benchmarks/pyrosetta_relax_validate.py` (audit) and `scripts/benchmarks/make_` (figure)

Alphabet expressivity ceiling (axis12/pair72/triple232 oracle)

- Reports: `artifacts/reports/step1_theta_star_step2_triple232_evidence_pack.md`, `artifacts/reports/pdb_oracle_codec_bench_n50_seed0.md` (and related)
- Primary figures: `paper/assets/figures/fig8.png` and `figures/triple232_oracle_codec_tm_bar.pn`
- Large-scale PDB extension: `artifacts/reports/pdb_oracle_codec_bench_n200_seed0_dirscale.md` and `figures/pdb_bench/oracle_codec_tm_hist_n200_seed0_dirscale.png`, `figures/pdb_bench/or`

Large-scale PDB phason proxy statistics (window sanity check)

- Reports: `artifacts/reports/pdb_phason_stats_n1000_seed0_*.md` (exact tag depends on null configuration)
- Samples CSV (long-form): `data/processed/pdb_bench/phason_stats_samples_n1000_seed0_*.csv`
- Summary CSV (per-protein): `data/processed/pdb_bench/phason_stats_summary_n1000_seed0_*.cs`
- Figures: `figures/pdb_bench/phason_phrms_box_triple232_n1000_seed0_triple232.png`, `figures/pdb_bench/phason_phrms_ecdf_triple232_n1000_seed0_triple232.png`, `figures/pdb_ben`
- Piecewise slice extension (seg=64): `artifacts/reports/pdb_phason_stats_n1000_seed0_triple232_p` `data/processed/pdb_bench/phason_stats_summary_n1000_seed0_triple232_piecewise64.csv`, and corresponding figures under `figures/pdb_bench/` matching `*piecewise64*.png`
- Satisfiability vs discriminability note (negative result summary): `artifacts/reports/phason_satisfiab`

Multi-scale symbolic certificate metrics (Fold_m on binary readouts)

- Protocol: define a binary readout $\rho(t) \in \{0,1\}$ and compute sliding windows $\omega_t^{(m)}$ of length m; stabilize each window via $\mathrm{Fold}_m : \{0,1\}^m \twoheadrightarrow X_m^{\mathrm{Z}}$ (no adjacent 11), producing a stable-type stream $x_t \in X_m^{\mathrm{Z}}$.
- Readouts used in this repository:
  - $\rho_A(t)$ from perp-space: with $w_0 = \mathrm{mean}(y^{\perp})$ and fixed $u = (1,0,0)$, define $s_t = u \cdot (y_t^{\perp} - w_0)$ and binarize via a robust threshold (default $b_t = 1\{s_t \geq \mathrm{median}(s)\}$; ablations use mean or zero threshold).
  - $\rho_B(t)$ from 6D steps: from $\Delta n_t = n_{t+1} - n_t$, binarize by majority sign across coordinates (deterministic tie-break).
- Metrics: empirical entropy of stable types $H(\text{type})$, support size, run-length statistics on the raw bitstream, and a practical SMB-style entropy-rate proxy (block-entropy slope).
- Null families: in addition to random-chain and perturbed-native controls, we include bond-direction shuffle and block-shuffle controls (preserve per-bond lengths and marginal directions while destroying long-range correlation; configured via `--shuffle-reps`, `--blockshuffle-reps`, `--blockshuffle-k`).
- Hybrid full1k run (recommended audit entrypoint): run directory and plots are provided in the evidence repository under `docs/runs/`.
- Outputs (large CSVs are under `data/processed/pdb_bench/` and reports under `artifacts/reports/`):

Table 6: Hard-negative decoy replay summary for ρ_A at $m = 8$ using $H(\text{type})$. "Native Rank" is the native percentile among decoys (lower is more ordered / more native-like). Cliff's δ is computed for Shared-PCA only (native vs decoys).

Target ID	Decoy set	Native Rank (Shared-PCA)	Native Rank (Best-Native)	Cliff's δ (Shared-PCA)
1CTF	4state_reduced	54.3%	36.5%	+0.084
1R69	4state_reduced	3.0%	3.0%	-0.970
1SN3	4state_reduced	28.5%	28.5%	-0.432
2CRO	4state_reduced	1.6%	1.6%	-0.984
3ICB	4state_reduced	17.0%	17.0%	-0.661
4PTI	4state_reduced	15.9%	3.2%	-0.688
4RXN	4state_reduced	17.1%	17.1%	-0.663
1CTF	LMDS	55.6%	38.5%	+0.111
1DTK	LMDS	5.1%	5.1%	-0.917
1FC2	LMDS	4.2%	4.2%	-0.916
1IGD	LMDS	25.0%	25.0%	-0.503
1SHF-A	LMDS	28.9%	23.6%	-0.422
2CRO	LMDS	1.8%	1.8%	-0.982
2OVO	LMDS	5.5%	0.9%	-0.899
4PTI	LMDS	15.7%	3.8%	-0.698
UNK	LMDS	14.4%	14.4%	-0.711

  - Report: `artifacts/reports/pdb_*_stats_n1000_seed0_*.md` (exact tag depends on null configuration)
  - Samples CSV: `data/processed/pdb_bench/*_stats_samples_n1000_seed0_*.csv`
  - Summary CSV: `data/processed/pdb_bench/*_stats_summary_n1000_seed0_*.csv`
  - Figures: generated by plotting scripts under `scripts/pdb_bench/`.

Hard-negative decoy benchmark (Decoys 'R' Us)

- Run directory, manifests, and plots are provided in the evidence repository under `docs/runs/`.
- Scripts: `scripts/pdb_bench/decoys_r_us_download.py` and certificate evaluation scripts under `scripts/pdb_bench/`.

Decoy replay robustness table (Shared-PCA vs Best-Native-Axis)

Toy decoy smoke (pipeline sanity check)

- Run directory: `docs/runs/` (toy native-vs-decoy smoke set)
- Script: `scripts/pdb_bench/toy_decoy_smoke.py`

Physical consistency audit (CA-level physics audit layer)

- Run directory (CSV + README): `docs/runs/physics_audit/`
- CSV (4state_reduced full audit): provided in `docs/runs/physics_audit/`
- Scripts: `scripts/audit/` (physics proxies + manifest audit + figure generation)
- Paper figure asset: Fig. S17

Table 7: Template/fragment dependence and cross-method topology agreement for QUARK vs I-TASSER with strict homology exclusions. TM-scores are computed by TM-align/US-align on model1 (unless stated). "A vs B" within-method TM indicates sensitivity to homology exclusion; "QUARK-A vs I-TASSER-A" indicates cross-method agreement. Final values are recorded in `docs/runs/quark_itasser_homology_ablation/tmalign_summary.csv`.

Target ID	QUARK-A vs QUARK-B (TM)	I-TASSER-A vs I-TASSER-B (TM)	QUARK-A vs I-TASSER-A (TM)	Notes
1R69	–	–	–	
2CRO	–	–	–	
4PTI	–	–	–	
1CTF	–	–	–	
1DTK	–	–	–	
1SHF-A	–	–	–	

QUARK vs I-TASSER homology-ablation benchmark (external baseline)

- Run directory (inputs + job tracking + summary CSV template): `docs/runs/quark_itasser_homology_`
- FASTA inputs: `docs/runs/quark_itasser_homology_ablation/targets_6.fasta`
- Job tracker: `docs/runs/quark_itasser_homology_ablation/targets.md`
- Server summary CSV (filled after downloads): `docs/runs/quark_itasser_homology_ablation/server_`
- TM-align record: `docs/runs/quark_itasser_homology_ablation/tmalign_summary.csv`
- Scripts: `scripts/benchmarks/extract_fasta_from_pdb_cache.py`, `scripts/benchmarks/parse_quar` `scripts/benchmarks/record_tmalign_results.py`
- TM-score note: `server_summaries.csv` reports a **Kabsch single-pass TM-score** computed under 1:1 residue correspondence (in-repo `scripts/pdb_bench/bench_utils.py`; source=`tm_kabsch`). This is *not* TM-align/US-align (no alignment search). Where an alignment-search TM is needed, we report it separately in `tmalign_summary.csv`. Fig. 7A uses the Kabsch single-pass TM-score for consistency across targets and methods under the same indexing convention.

Blind sprint + certificate prototypes (small N)

- Run directory and reports: provided in the evidence repository under `docs/runs/`
- Scripts: `scripts/blind_sprint/` (blind sprint runners and suite scripts)

AlphaFold vs PhasonFold transparency figure (additional AFDB example)

- Primary composite figure (main text): `paper/assets/figures/fig_af_vs_omega.png`
- Auditable run folder: provided in the evidence repository under `docs/runs/`
- Supplementary robustness panel (AFDB P04637; mixed confidence): `paper/assets/figures/fig_af_vs`

Large-scale PDB chain-consistent 6D beam-lift benchmark

- Report: `artifacts/reports/pdb_beam_lift_bench_n1000_seed0_pair72_beam8.md`
- Summary CSV: `data/processed/pdb_bench/beam_lift_bench_summary_n1000_seed0_pair72_beam8`

- Certificate summary CSV (if enabled): `data/processed/pdb_bench/beam_lift_auric_summary_n1000_`
- Figures: `figures/pdb_bench/beam_lift_phrms_box_n1000_seed0_pair72_beam8.png`, `figures/pdb_bench/beam_lift_phrms_ecdf_n1000_seed0_pair72_beam8.png`, `figures/pdb_bench/b`
- High-end configuration (triple232 + piecewise seg=64): `artifacts/reports/pdb_beam_lift_bench_n10` `data/processed/pdb_bench/beam_lift_bench_summary_n1000_seed0_triple232_beam8_pw64.csv`, and figures `figures/pdb_bench/beam_lift_*_n1000_seed0_triple232_beam8_pw64.png`

Sprint-to-0.9 A/B (1TIM & 2O5P tail250)

- Oracle-guided compilation runner report: `artifacts/reports/sprint0p9_AB_runner_report.md`
- Blind Sprint report: `artifacts/reports/blind_sprint_AB_report_v1.md`
- CSV summary used by paper tables: `data/processed/sprint_to_0p9_AB_summary.csv` (Blind Sprint A/B) and `data/processed/sprint_to_0p9_ab_runs.csv` (per-seed)
- Figures: Fig. 11–Fig. 14 (oracle sprint) and Fig. 15 (Blind Sprint)

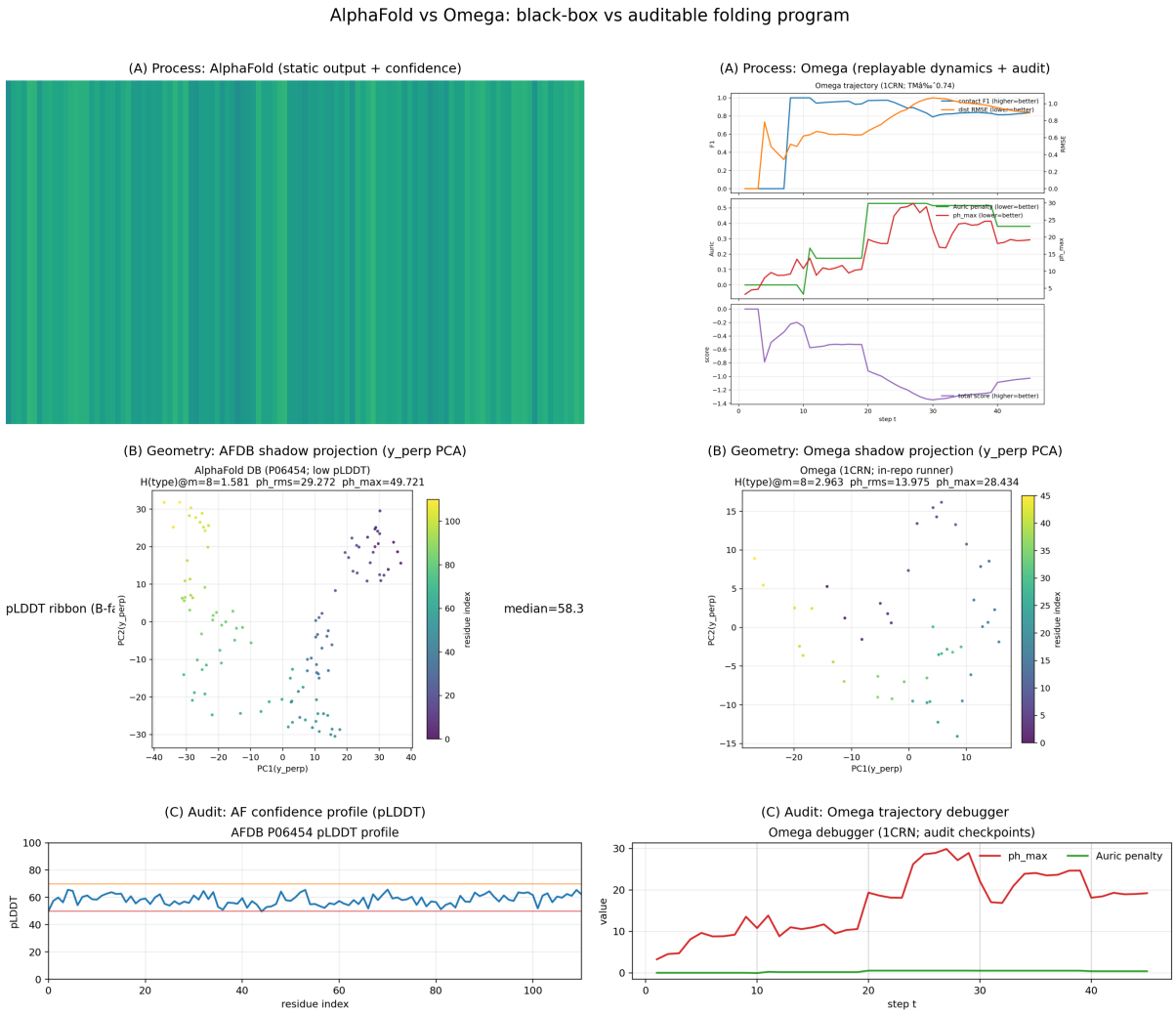

Figure 16: **AlphaFold vs PhasonFold: black-box output vs replayable, auditable dynamics. (A) Process:** AlphaFold DB provides a static structure estimate and a per-residue confidence profile (pLDDT), while PhasonFold exposes an explicit, replayable trajectory with logged objective terms and audit readouts. **(B) Geometry:** a "shadow projection" of the perpendicular-space trace ($y_\perp$ PCA) highlights a qualitative difference between an AFDB low-confidence intrinsically disordered example (UniProt P06454) and a PhasonFold reconstruction example (1CRN) under the same 6D lifting/quantization protocol. **(C) Audit/debugger:** AlphaFold exposes confidence as an output-side annotation, whereas PhasonFold exposes an internal audit signal (phason proxy and checkpoint diagnostics) that supports targeted interventions and operator-level A/B tests. All run artifacts used to build this figure are provided in the accompanying evidence repository. As a robustness check against "hand-picked IDP" criticism, we also include a second AFDB example with mixed-confidence regions (UniProt P04637; p53).

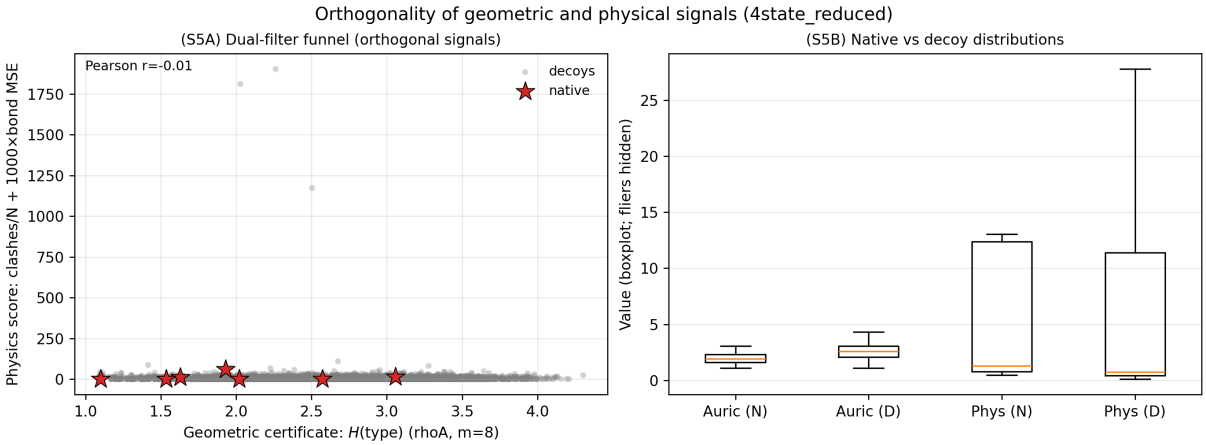

Figure 17: **Orthogonality of geometric and physical signals (post-hoc; CA-level).** We compare a geometry-based certificate (
rho_A type entropy at $m = 8$; x-axis) against a normalized physics score derived from two C
$alpha$ proxies (y-axis): clashes/N + 1000
$times$bond-length MSE. On the hard-negative decoy benchmark `4state_reduced`, these signals are weakly correlated (Pearson r
$approx$0), yielding a dual-filter interpretation: many decoys satisfy local physical constraints while violating global geometric manifold constraints (high geometric type entropy), while a smaller subset exhibits the converse.